A Meta-Law to Rule Them All: Physicists Devise a "Theory of Everything"

"Constructor theory" unites in one framework how information is processed in the classical and quantum realms

May 26, 2014 | By Zeeya Merali

"Once you have eliminated the impossible," the fictional detective Sherlock Holmes famously opined, "whatever remains, however improbable, must be the truth." That adage forms the foundational principle of "constructor theory"—a candidate "theory of everything" first sketched out by David Deutsch, a quantum physicist at the University of Oxford, in 2012. His aim was to find a framework that could encompass all physical theories by determining a set of overarching "meta-laws" that describe what can happen in the universe and what is forbidden. In a May 23 paper posted to the physics preprint server, arXiv, constructor theory claims its first success toward that goal by unifying the two separate theories that are currently used to describe information processing in macroscopic, classical systems as well as in subatomic, quantum objects.

Painted portrait of Claude Shannon, the "father of information theory".
Credit: Flickr/thierry ehrmann

Computer scientists currently use a theory developed by the American mathematician and cryptographer Claude Shannon in the 1940s to describe how classical information can be encoded and transmitted across noisy channels efficiently—what, for instance, is the most data that can be streamed, in theory, down a fiber-optic cable without becoming irretrievably corrupted. At the same time, physicists are striving to build quantum computers that could, in principle, exploit peculiar aspects of the subatomic realm to perform certain tasks at a far faster rate than today's classical machines.

But the principles defined by Shannon's theory cannot be applied to information processing by quantum computers. In fact, Deutsch notes, physicists have no clear definition for what "quantum information" even is or how it relates to classical information. "If we want to make progress in finding new algorithms for quantum computers, we need to understand what quantum information actually is!" he says. "So far, the algorithms that have been discovered for quantum computers have been surprises that were discovered by blundering about because we have no underlying theory to guide us."

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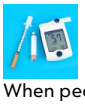

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In 2012 Deutsch outlined constructor theory, which, he believes, could provide the underlying foundation for a grand unification of all theories in both the classical and quantum domains. This latest paper is a first step toward that larger goal—a demonstration of how classical and quantum information can be used to unify two physical theories. According to constructor theory, the most fundamental components of reality are entities—"constructors"—that perform particular tasks, accompanied by a set of laws that define which tasks are actually possible for a constructor to carry out. For instance, a kettle with a power supply can serve as a constructor that can perform the task of heating water. "The language of constructor theory gives a natural way to describe the most fundamental principles that must be obeyed by all subsidiary theories, like conservation of energy," explains Chiara Marletto, a quantum physicist also at Oxford, who co-authored the new paper. "You simply say that the task of creating energy from nothing is impossible."

Dean Rickles, a philosopher of physics at the University of Sydney who was not involved in the development of the theory, is intrigued by its potential to unify nature's laws. "It's a very curious new take on the notion of a theory of everything," he says. "In principle, everything possible in our universe could be written down in a big book consisting of nothing but tasks [and in] this big book will also be encoded all of the laws of physics."

To develop their description of information, Deutsch and Marletto homed in on one key task that is possible in classical systems but impossible in quantum systems: the ability to make a copy. Since the 1980s physicists have known that it is impossible to make an identical copy of an unknown quantum state. In their new paper Deutsch and Marletto define a classical information medium as one in which states can all be precisely copied. They then work out which tasks must be possible in such a system to remain in line with Shannon's theory.

The collaborators then go on to define the concept of a "superinformation" medium that encodes messages that specify particular physical states—in this case, one in which copying is impossible. They discovered that a special subset of their superinformation media display the properties associated with quantum information processing. "We found that with this one constraint in place telling you what you cannot do in a superinformation medium—the task of copying—you end up discovering the weird new information-processing power that is a property of quantum systems," Marletto says.

The team showed that with this restriction on copying in place a number of other properties begin to emerge: Measuring the state of a superinformation medium will inevitably disturb it—a feature commonly associated with quantum systems. But because it is forbidden to make an exact copy of certain sets of states in a superinformation medium this forces some uncertainty into the outcome of the measurement.

The team has also shown that entanglement—the spooky property that binds quantum objects together so that they act in tandem, no matter how far apart they are—also arises naturally, once this constraint on copying is in place. According to Marletto, the crucial property of a system containing two entangled states is that the information stored in the combined system is more than the information that can be gleaned just by examining each member of the pair individually. The quantum whole is

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more than the sum of its parts.

In their paper Deutsch and Marletto demonstrate that information can be encoded in two superinformation media in such a way that it is impossible to retrieve it by measuring the single subsystems separately—that is, entanglement is inevitable. Similarly, in a classical medium, entanglement is impossible. “The appealing thing about this formalism is the way that common features of quantum mechanics fall out,” says Patrick Hayden, a quantum physicist at Stanford University, adding: “I have real respect for the creative thinking behind constructor theory and its ambitions.” He notes, however, that there are competing attempts by other researchers to develop a deeper understanding of quantum mechanics, including ideas based on copying, and as yet it is too early to say which, if any, will prove to be the best description.

Rickles agrees that it will take time for physicists to verify that the theory—which has not yet passed through peer review—is truly successful at uniting classical and quantum information theory. But if affirmed, it would give a boost to Deutsch’s goal to help in the hunt for the long-sought theory of quantum gravity, uniting the currently incompatible quantum theory and general relativity. “This is the first time in the history of science that it’s known that our deepest theories are wrong, so it’s obvious that we need a deeper theory,” Deutsch says.

Rickles believes that constructor theory has the potential to prescribe meta-laws that general relativity and quantum theory must obey. “The meta-laws are more stable creatures, they survive scientific revolutions,” he says. “Having such principles in hand gives us a better grasp of the nature of reality. I’d say that’s a pretty good advantage.”

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jtdwyer May 26, 2014, 9:57 AM

At first reading, this appears to be just more imaginary information as it pertains to physical systems...

What medium - what information - what physical processes transcribe information into physical systems and encode information from physical systems?

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dracma May 26, 2014, 12:13 PM

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If so why not a Nobel Award for this fundamental research?

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jtdwyer May 26, 2014, 8:54 PM

I reiterate - utter nonsense! Information is in the mind of the observer. In physical systems, information produces no physical effects - there are no physical processes that produce physical effects based on information. Information in physics is merely an artifact of the observation of physical properties. Without observers no information exists as a discrete physical element. As